

NAME: _____

PERIOD: _____

A Test of Mean Resolution Time

AP Statistics · Unit 4 · Topic 4.5

[STAR] THE PROBLEM

Suppose a help desk closed 1,200 tickets in a quarter. To investigate whether their population mean resolution time was below 12 hours, an analyst took an SRS of 24 and used $\alpha = 0.05$.

Leah: “The lower-tail test gives $t = -1.5747$, $df = 23$, and $p \approx 0.0645$. I would state what evidence this does and does not provide.”

Damon: “Because $p > 0.05$, accept H_0 . The test proves the population mean was exactly 12 hours and the observed difference was only chance.”

Ticket sample

Summary	Value
Population	1200
n	24
Mean (h)	11.1
SD (h)	2.8
Shape	Symmetric; no outliers

FIRST TAKE — one sentence before you work the parts

Does this small study seem able to prove that the average was exactly 12 hours? Explain in one sentence.

[BOOK] COMPARE EVIDENCE TO THE PRESET THRESHOLD (keep on the page)

For a one-sample mean test with unknown population standard deviation, $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$ has $df = n - 1$. The p -value is the probability, assuming H_0 is true, of a test statistic at least as extreme as the observed statistic in the direction of H_a . If $p \leq \alpha$, reject H_0 ; if $p > \alpha$, fail to reject H_0 . Failing to reject says the sample does not provide convincing evidence for H_a at that significance level. It does not prove the null value, and the conclusion must concern the population represented by the random sample.

Work (a)–(c).

DOK 1 (a) State H_0 and H_a for the population mean resolution time.

DOK 2 (b) Using $\bar{x} = 11.1$, $s = 2.8$, and $n = 24$, calculate t . Use the supplied $p = 0.0645$; you do not need to find it again.

DOK 3 (c) In 3–4 sentences, decide whose account is justified. Compare the p-value with 0.05, state the conclusion for all 1,200 tickets, and explain why failure to reject does not prove equality or that chance was the only explanation.

Frame: The numerical comparison is p _____ α , so the formal decision is _____ and the population conclusion is _____.

Frame: That decision _____ establish _____ because the p-value is calculated assuming _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored **E / P / I**.